Standard / Generation	Year Ratified	Frequency Band	PHY Standard	Max Data Rate	Modulation Techniques
802.11 Prime	1997	2.4 GHz ISM	DSSS, FHSS, IR	2 Mbps	Barker Code
802.11b	1999	2.4 GHz ISM	HR-DSSS	11 Mbps	CCK
802.11a	1999	5 GHz U-NII	OFDM	54 Mbps	BPSK, QPSK, 16-QAM, 64-QAM
802.11g	2003	2.4 GHz ISM	ERP (ERP-OFDM & ERP-DSSS)	54 Mbps	OFDM (Up to 64-QAM), CCK
802.11n (Wi-Fi 4)	2009	2.4 GHz & 5 GHz	HT (MIMO + OFDM)	600 Mbps	Up to 64-QAM
802.11ac (Wi-Fi 5)	2013	5 GHz (below 6 GHz)	VHT (MU-MIMO)	6933.3 Mbps	256-QAM
802.11ad (WiGig)	2012	60 GHz	DMG	7 Gbps	SQPSK, QPSK, 16-QAM, 64-QAM
802.11af	2014	TV White Space (54–790 MHz)	TVHT (OFDM based)	568 Mbps	Up to 256-QAM
802.11ah (HaLow)	2016	Sub-1 GHz	Wi-Fi HaLow	~347 Mbps	BPSK, QPSK, up to 256-QAM
802.11ax (Wi-Fi 6/6E)	2021	2.4 GHz, 5 GHz, 6 GHz	HE (OFDMA)	9.6 Gbps	Up to 1024-QAM
802.11be (Wi-Fi 7)	2024	2.4 GHz, 5 GHz, 6 GHz	EHT	46 Gbps	Up to 4096-QAM
802.11bn (Wi-Fi 8)	*Draft (Est. 2028)*	2.4 GHz, 5 GHz, 6 GHz	UHR (Ultra High Reliability)	46 Gbps	Up to 4096-QAM

U-NII Band	Frequency Range	Bandwidth	20 MHz Channels	Key Characteristics & Restrictions
U-NII-1 (Lower)	5.150 GHz – 5.250 GHz	100 MHz	4 channels	Originally restricted to indoor use in the US, but the FCC lifted this restriction in 2014.
U-NII-2A (Middle)	5.250 GHz – 5.350 GHz	100 MHz	4 channels	Radios transmitting in this band must support Dynamic Frequency Selection (DFS) to avoid radar interference.
U-NII-2C (Extended)	5.470 GHz – 5.725 GHz	255 MHz	12 channels	Requires DFS. Includes Channel 144, which was added with the 802.11ac amendment.
U-NII-3 (Upper)	5.725 GHz – 5.850 GHz	125 MHz	5 channels	Expanded by the FCC in 2014 to include Channel 165. Often used for outdoor point-to-point links.
U-NII-4	5.850 GHz – 5.925 GHz	75 MHz	(Pending)	Historically reserved for Intelligent Transportation Systems (ITS) and DSRC, but the FCC has proposed repurposing the lower 45 MHz for Wi-Fi.
U-NII-5 (6 GHz)	5.925 GHz – 6.425 GHz	500 MHz	Part of 59 new US channels	Indoor low-power APs allowed (30 dBm EIRP max). Outdoor standard-power APs (36 dBm EIRP max) allowed but **require Automated Frequency Coordination (AFC)** to protect incumbents.
U-NII-6 (6 GHz)	6.425 GHz – 6.525 GHz	100 MHz	Part of 59 new US channels	Indoor low-power APs allowed. **No outdoor Wi-Fi permitted** due to mobile and fixed satellite service incumbents.
U-NII-7 (6 GHz)	6.525 GHz – 6.875 GHz	350 MHz	Part of 59 new US channels	Indoor low-power APs allowed. Outdoor standard-power APs allowed but **require AFC**.
U-NII-8 (6 GHz)	6.875 GHz – 7.125 GHz	250 MHz	Part of 59 new US channels	Indoor low-power APs allowed. **No outdoor Wi-Fi permitted** due to broadcast and fixed satellite service incumbents.